MILITARY SPECIFICATION SHEET

RECEIVER, RANGE SAFETY, COMMAND
TYPE II, CLASS 1, 2, 3, STYLE C1, C2, C3, C4, C5

This specification is approved for use by the US Army Test and Evaluation Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the receiver described herein shall consist of this document and the latest issue of Specification MIL-R-87954(TE).

REQUIREMENTS:

Electrical: (all classes unless otherwise specified)

Operating frequency (3.5.1.2) and (6.2) specified in ordering data.

RF input level monitor (3.5.1.3):

- DC voltage range: 0 to 4.0 volts or greater, 5.0 volts maximum
- Output impedance: less than 5K ohms
- Rate of change: 0.02 volt/db minimum
- RF level range: -107 to -80 dbm (min)
- Repeatability: ±2db when calibrated at a given temperature from -40°C to 71°C
- Output connection: Specified by Style

Predetection bandwidth (3.8.1):

- Reference level: -107 dbm
- 3 db bandwidth: 180 KHz minimum
- 60 db bandwidth: 360 KHz maximum

Decoder command channels (3.5.2):

- Total no: Specified by Style
- Frequency: Specified in ordering data (6.2)

Output section (3.5.3):

Class 1 Output relays: MIL-R-39016/13 (or other equivalent type(s) as specified in ordering data). Number of relays and command function logic specified by Style.

Class 2 Solid state output:

Low level (source)

- Output activated: 14.0 vdc ≥ output voltage ≥ 12 vdc
- Output deactivated: 2.5 vdc ≥ output voltage ≥ 0
- Output resistance: ≤100 ohms
- Load resistance: ≥3 kiloohms
- Number of outputs and command function logic specified by style.

Medium power level (source):

- Activated: Current output - 0.50A min. continuous
  From aux power voltage source - 24 to 34 vdc
  Max voltage drop aux power voltage to output with current @0.50A - 1.5 vdc
  Receiver power voltage range - 22 to 34 vdc
- Not activated: High impedance 1 meg ohm min. isolation to any power source and common.
- Number of outputs: Specified by Style

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Function logic: Output(s) to function in accordance with the Function Table of the particular style.
Class 3 Contains combination of semiconductor and relay outputs. Number of outputs and function logic specified by style.

Power control circuit (3.5.4):
Magnetic latch relay: MIL-R-39016/31 (or other equivalent type(s) as specified in ordering data). Specified by style when required.

Prime power (3.8.11.1):
Nominal voltage: 28 vdc
Voltage range: 22-35 vdc

Input current (3.8.11.2): 180 ma max @ 28 vdc (with no outputs activated)
Output activation current: 30 ma maximum per output activation

Performance characteristics:
RF signal level range (3.8.5): +13 dbm to ≤107 dbm
Threshold sensitivity (3.8.2): ≤-107 dbm with permissible drop out rate of one per channel per five seconds
Response time (3.8.10): 25 milliseconds or less at RF input level ≥-95 dbm

Electromagnetic interference: (all classes)
Spurious response rejection (antenna) (3.10.1):

	CS 04	CS 08
Reference level:	-107 dbm	-107 dbm
Limit A:	-30 dbm*	-30 dbm*
Limit B:	0 dbm	0 dbm

*Limit for half IF image -50 dbm
IF image(s) - 40 dbm

Spurious response (power lead) (3.10.2)
Transient levels: ≤40 volts
Duration(s): ≤2 seconds

Environmental (3.9): (all classes)
Acceleration level (3.9.1): 100g
Altitude (3.9.2):

	Condition I	Condition II
Temperature:	-40°C	70°C
Altitude:	≤.1 in. Hg	≤.1 in. Hg
Operating time:	1 hour	1 hour

Shock (3.9.3):

Level (g)	Duration (milliseconds)
150 (g)	1.0
100	10.0

Explosive Generated Shock Spectrum: Damping Factor = .005

From Hz	to Hz	level(g) (±3db)
4000	10000	5000
4000	50	-9db/octave
10000	above	-3db/octave

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Vibration (3.9.4)
Qualification:
Sine Curve S Exposure time: 5 min. each axis
Random (Level profile ±3 db):

Freq (Hz)	Level (g^2/Hz)	Level (g^2/Hz)	Level (g^2/Hz)
20	.024	.00054	.02
100	0.60	.0135	0.44
1000	0.60	.0135	0.44
2000	0.15	.0033	0.122
Approx G (RMS)	29.4	4.4	25.3
Exposure time (min):	3.4	8.1	1.5

Random (Level profile ±3 db):

Freq (Hz)	Level (g^2/Hz)
30	.001
110	0.32
300	0.32
1000	0.001
Approx G(RMS)	9.5
Exposure time:	50 minutes

Physical characteristics: (all classes)
Case (3.11.1): In accordance with Figure 1, Receiver Outline
Connectors (3.7.2):
Antenna (RF) connector: J-1 M39012
Single output connector configuration: J-2 M24308/3-2, mounted in accordance with Figure 1A
Double output connector configuration: J-2 MS 27499E10B99P, J-3 MS 27499E10B99SC, mounted in accordance with Figure 1A
Wiring: In accordance with style requirements. All relay contact terminals and solid state outputs shall be accessible for output wiring reconfiguration by removing cover plate(s) and the output connector.
Total weight (3.11.1): ≤.95 pounds
Style Requirements:
Style C1:
Decoder Command Channels (3.5.2)
Total number: 3 channels
Frequency: (6.2) Specified in ordering data
Output section (3.5.3)
Output relays: 3 ea Class 1 output relays, one for each command channels to function in accordance with Table 1A.
Channel status monitor (3.5.2.3): Input voltage/channel monitor, see Figure 2A. R_1, R_2, and R_3: 100K ohms unless otherwise specified in ordering data.
Wiring: Wiring configuration in accordance with Figure 2A. Unless otherwise specified in ordering data.
Power control circuit (3.5.4)
Magnetic latch relay: MIL-R-39016/31 (or other equivalent type(s) as specified in ordering data).

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TABLE 1A Function Table

Style C1

Command channel inputs channel designation			Relay operation channel		
1st	2nd	3rd	1st	2nd	3rd
0 1/	0	0	0 1/	0	0
0	0	1 2/	0	0	1 2/
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	0
1	1	1	1	1	1

1/ 0 = Off or not activated
2/ 1 = On or activated

Style C2:
Decoder command channels (3.5.2)
Total number: 3 channels
Frequency: (6.2) Specified in ordering data
Audio output (3.5.1.4):
RF input level: ≥-70 dbm
Modulation: ±30 KHz (+10%) peak deviation
Level: 0.5 volt RMS ±3 db
Frequency range: 7.5 KHz to 50 KHz
Load impedance: ≥3 kilohm
Output impedance: ≤100 ohm
Output connection: See Figure 2B
Output section (3.5.3):
Output relays: 3 each Class 1 output relays, one per channel to function in accordance with Table 1B.
Wiring: Wiring in accordance with Figure 2B unless otherwise specified in ordering data.
Connector: J-2 Type M24308/3-2
Channel status monitor (3.5.2.3):
Input voltage/channel monitor, see Figure 2B, R_1, R_2, and R_3: 100K ohms unless otherwise specified in ordering data.

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TABLE 1B Function Table

Style C2

Command channel inputs channel designation			Relay operation/ channel		
1st	2nd	3rd	1st	2nd	3rd
0 1/	0	0	0	0	0
0	0	1 2/	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	0
1	1	1	1	1	1

1/ 0 = Off or not activated
2/ 1 = On or activated

Style C3:
Decoder command channels (3.5.2)
Total No: 4 channels
Frequency: (6.2) Specified in ordering data.
Output section (3.5.3):
Output relays: 4 each Class 1 output relays (1 each per channel)
Solid state output: 4 each Class 2 solid state, low level outputs (1 each per channel)
Command/output function: In accordance with Table 1C.
Connector: J-2 Type M24308/3-2
Wiring: Wiring in accordance with Figure 2c.
Channel status monitor (3.5.2.3): Shall be provided for each channel by coupling to the low level solid state output with isolation resistors R_1, R_2, R_3, and R_4 (*see* Figure 2c). R_1, R_2, R_3, and R_4 resistor values shall be 100K ohm unless otherwise specified in the ordering data.

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TABLE 1C Function Table

Style C3

Command channel inputs channel designation				Output operation/ channel designation			
1st	2nd	3rd	4th	1st	2nd	3rd	4th
0 1/	0	0	0	0	0	0	0
0	0	0	1 2/	0	0	0	1
0	0	1	0	0	0	1	0
0	0	1	1	0	0	1	1
0	1	0	0	0	1	0	0
0	1	0	1	0	1	0	1
0	1	1	1	0	1	1	1
1	0	0	0	1	0	0	0
1	0	0	1	1	0	0	1
1	0	1	0	1	0	1	0
1	0	1	1	1	0	1	1
1	1	0	0	1	1	0	0
1	1	0	1	1	1	0	1
1	1	1	0	1	1	1	0
1	1	1	1	1	1	1	1

1/ 0 = Off or not activated
2/ 1 = On or activated

Style C4:
Decoder command channels (3.5.2):
Total No: 3 channels
Frequency: (6.2) Specified in ordering data.
Output section (3.5.3): Three (3) each Class 2 medium power level outputs functionally connected to 1st, 2nd, and 3rd command channels.
Command/Output function: In accordance with Table 1D.
Connectors: J-2 Type MS 27499E10B99P
J-3 Type MS 27499E10B99SC
Output wiring: In accordance with Figure 2D.
Channel status monitor (3.5.2.3): One each per channel.
Open circuit output voltage:
Channel activated: ≥12 vdc and ≤14 vdc
Channel not activated: ≥0 and ≤2.5 vdc
Output resistance: 10K ohms ±10% unless otherwise specified in ordering data.
Power Control Circuit (3.5.4)
Magnetic latch relay: MIL-R-39016/31 or other equivalent type(s) as specified in ordering data.

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TABLE 1D Function Table

Style C4

Command channel inputs channel designation			Receiver Output		
1st	2nd	3rd	A	B	C
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	0
1	1	1	1	1	1

0 = Channel not activated
1 = Channel activated

0 = Zero output (isolated)
1 = Source 28V nominal

Style C5:
Decoder command channels (3.5.2):
Total No: Four (4) channels
Frequency (6.2): Specified in ordering data
Output section (3.5.3): Four (4) ea class 1
Output relays, one per channel
Command/output function: In accordance with Table 1E
Connectors: J-2 Type MS27499E10B99P
J-3 Type MS27499E10B99SC
Output wiring: In accordance with Figure 2E

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TABLE 1E Function Table

Style C5

Command channel inputs channel designation				Output designation				
1st	2nd	3rd	4th	Check Mon	4th Mon	OC	Arm	Dest
0 1/	0	0	0	0	0	0	0	0
0	0	0	1 2/	0	1	0	0	0
0	0	1	0	1	0	0	0	0
0	0	1	1	1	1	0	0	0
0	1	0	0	0	0	0	0	0
0	1	0	1	0	1	1	0	0
0	1	1	1	1	1	1	0	0
1	0	0	0	0	0	0	0	0
1	0	0	1	0	1	0	1	0
1	0	1	0	1	0	0	0	0
1	0	1	1	1	1	0	1	0
1	1	0	0	0	0	0	0	0
1	1	0	1	0	1	0	0	0
1	1	1	0	1	0	0	0	0
1	1	1	1	1	1	0	1	0
LOGIC SEQUENCE (activate in order listed)								
1	0	0	1	0	1	0	1	0
1	0	0	0	0	0	0	1	0
1	1	0	0	0	0	0	1	1

1/ 0 = Off or not activated
2/ 1 = On or activated

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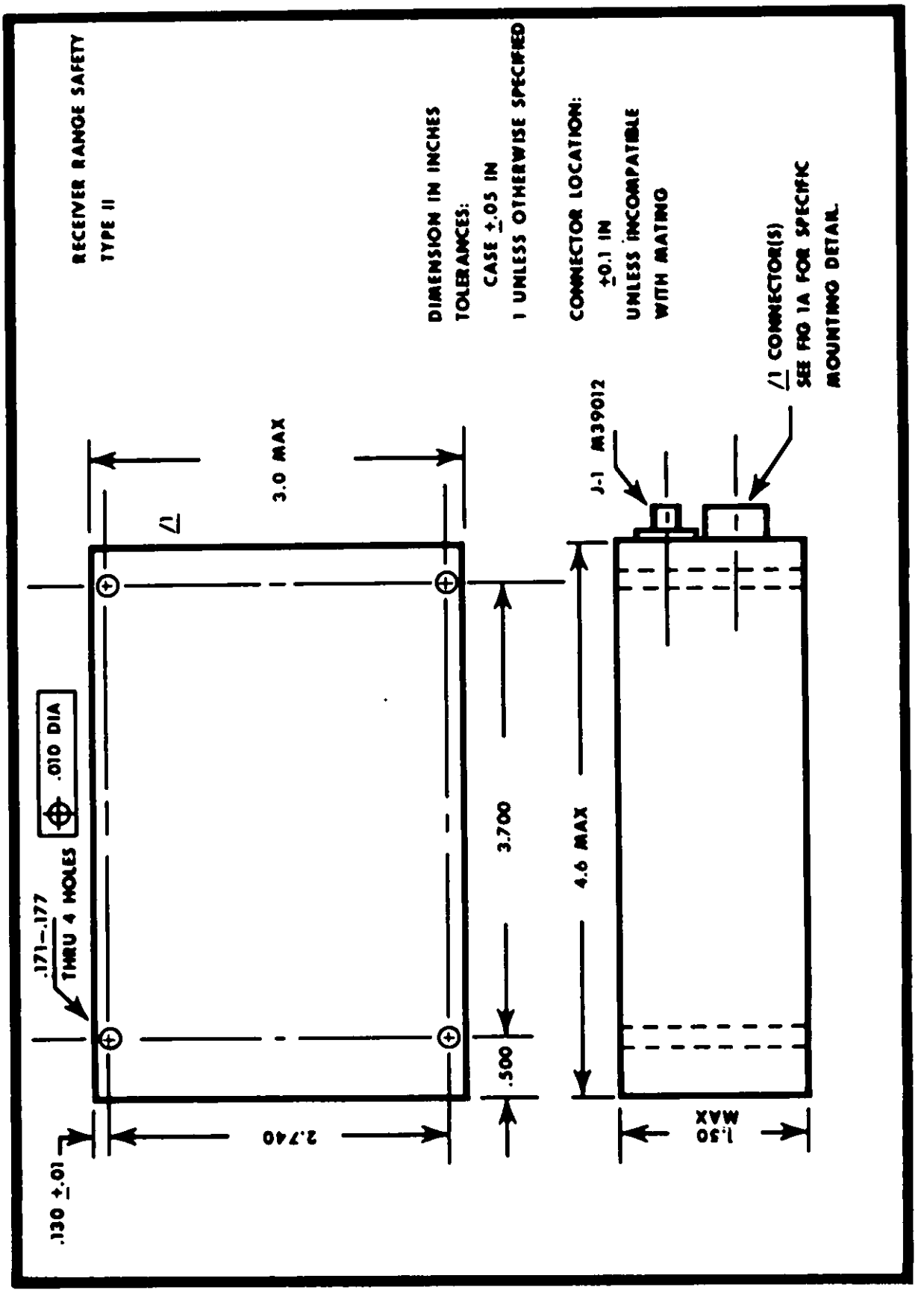


FIGURE 1 RECEIVER OUTLINE

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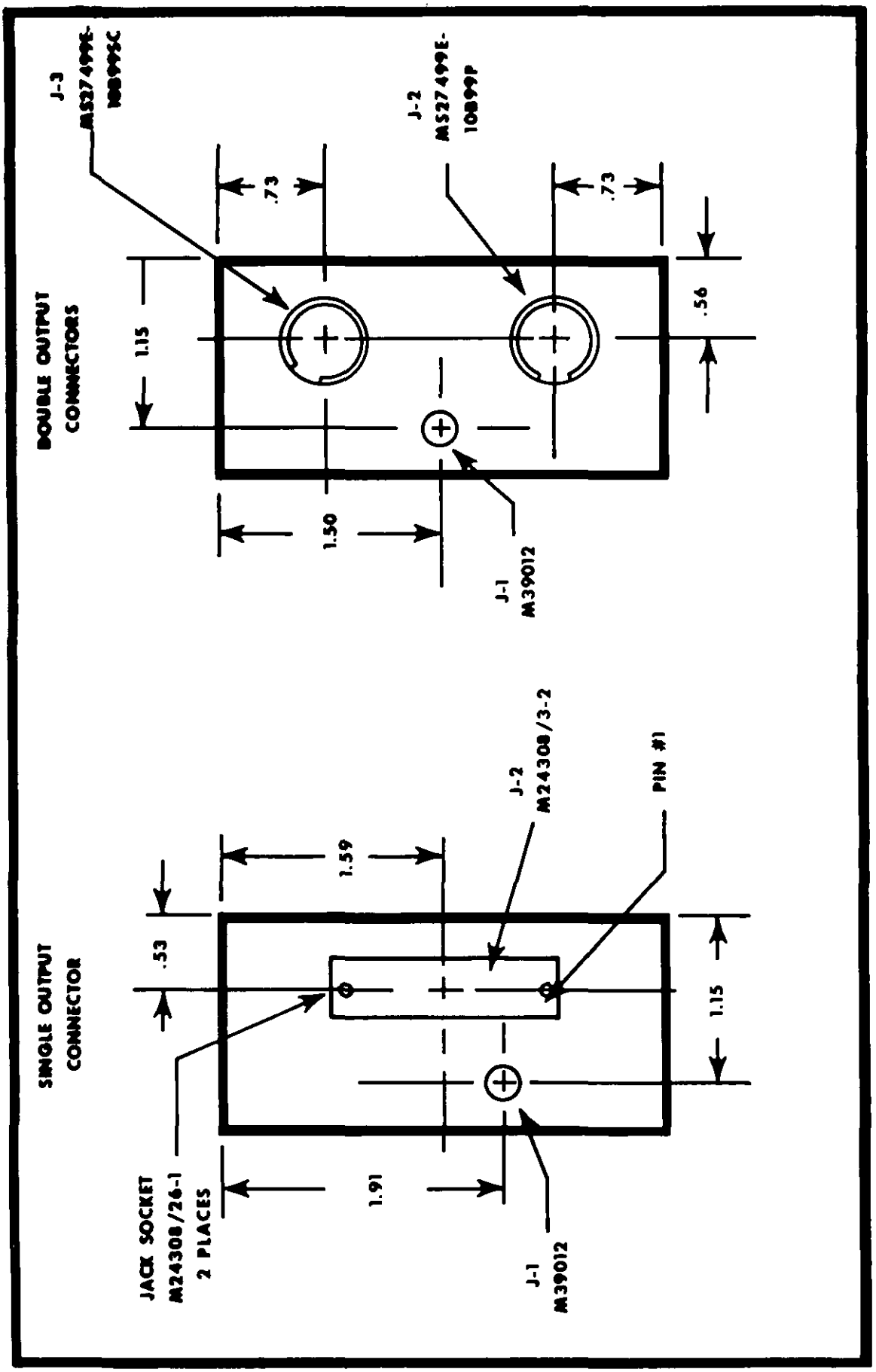


FIGURE 1A CONNECTOR MOUNTING DETAIL
SINGLE AND DOUBLE CONFIGURATIONS

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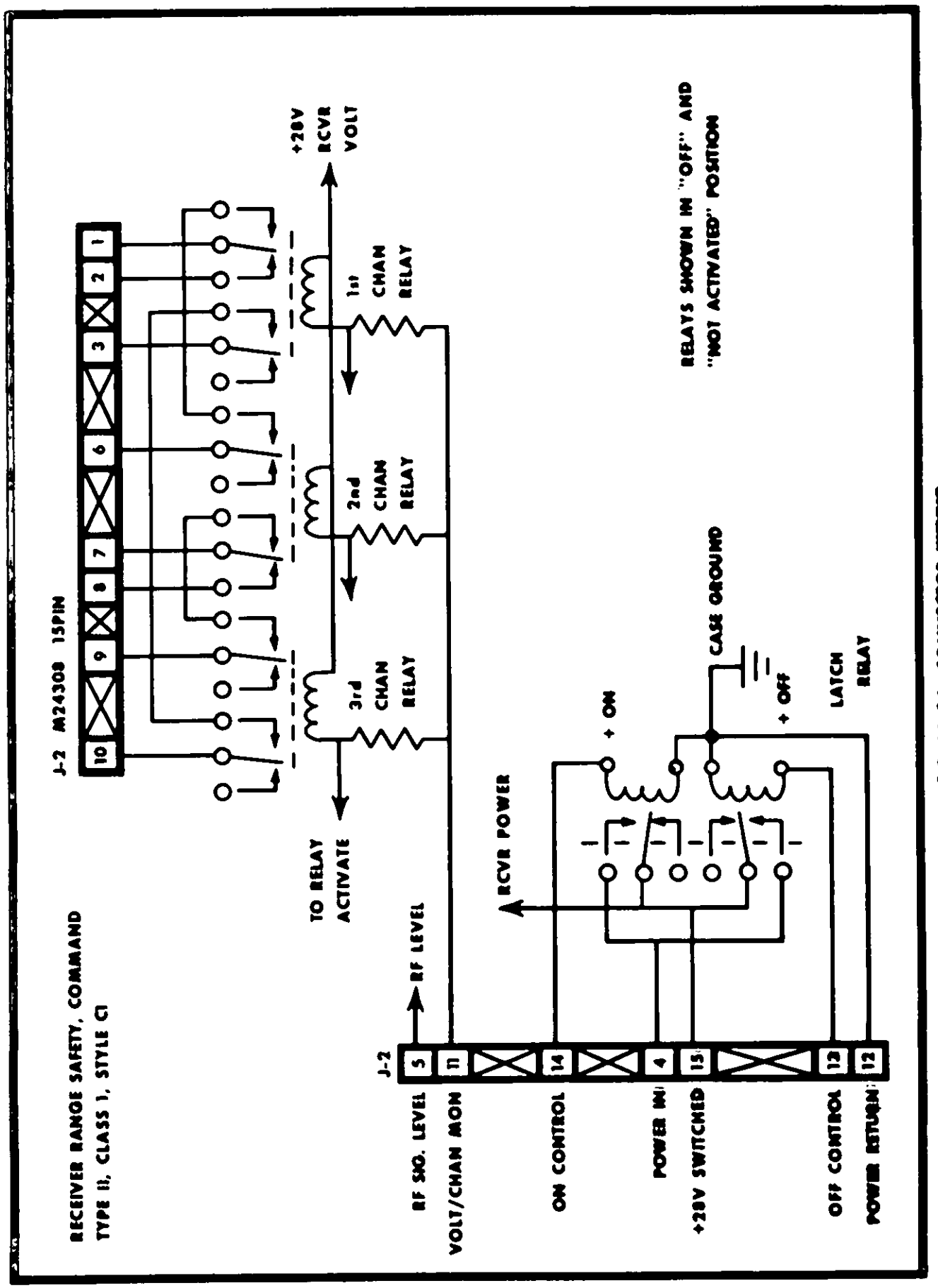


FIGURE 2A CONNECTOR WIRING

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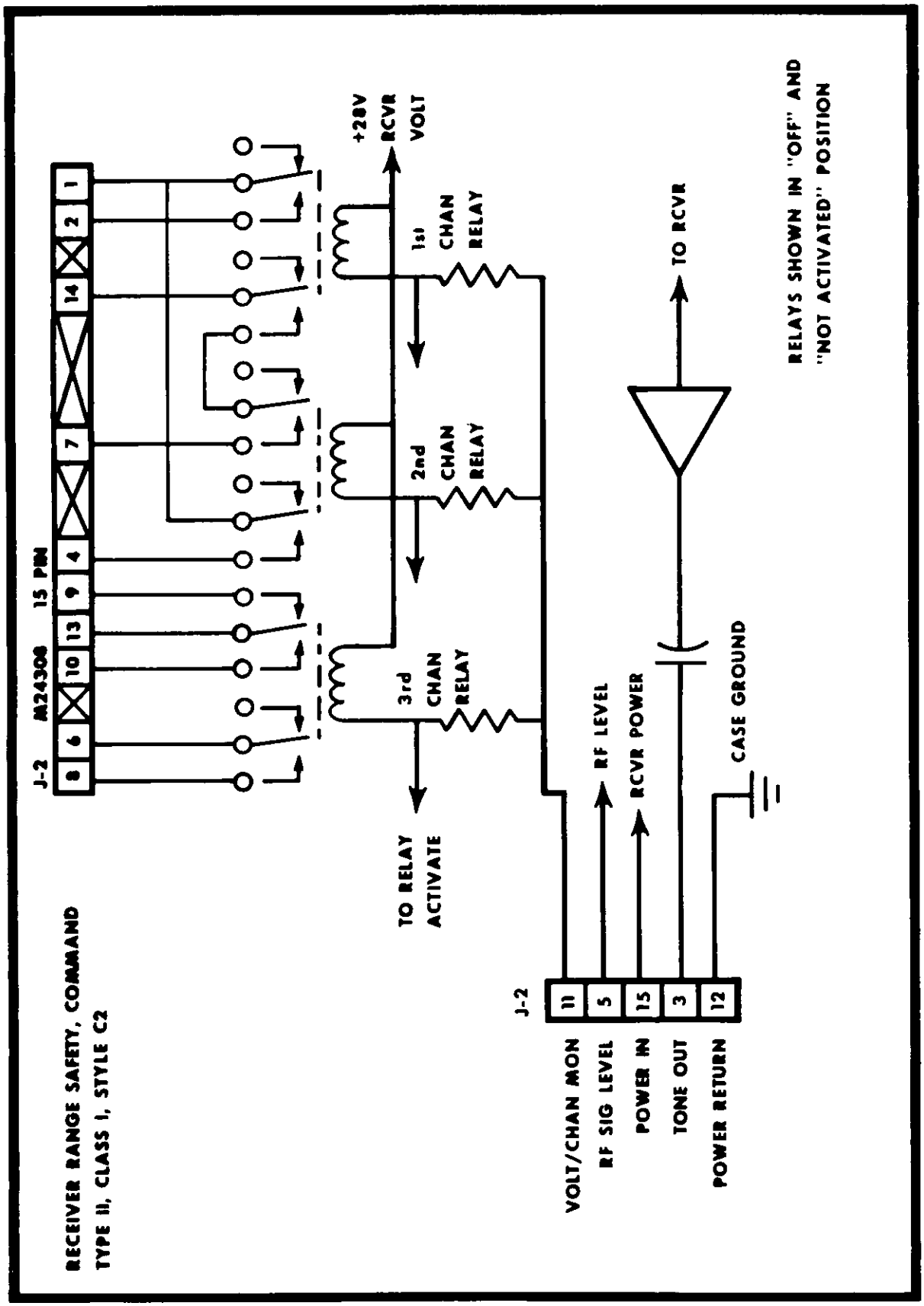


FIGURE 2B CONNECTOR WIRING

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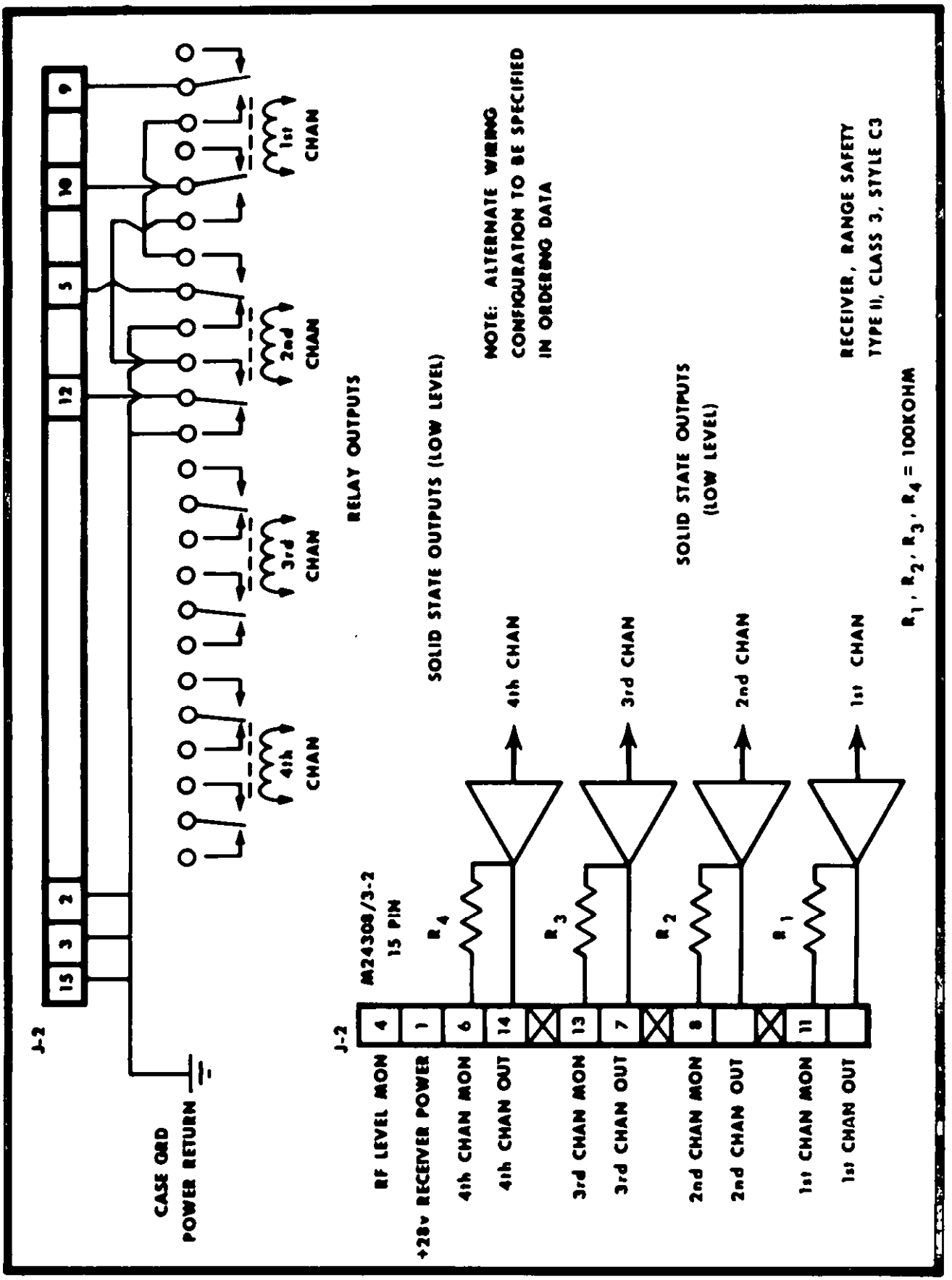


FIGURE 2C CONNECTOR WIRING

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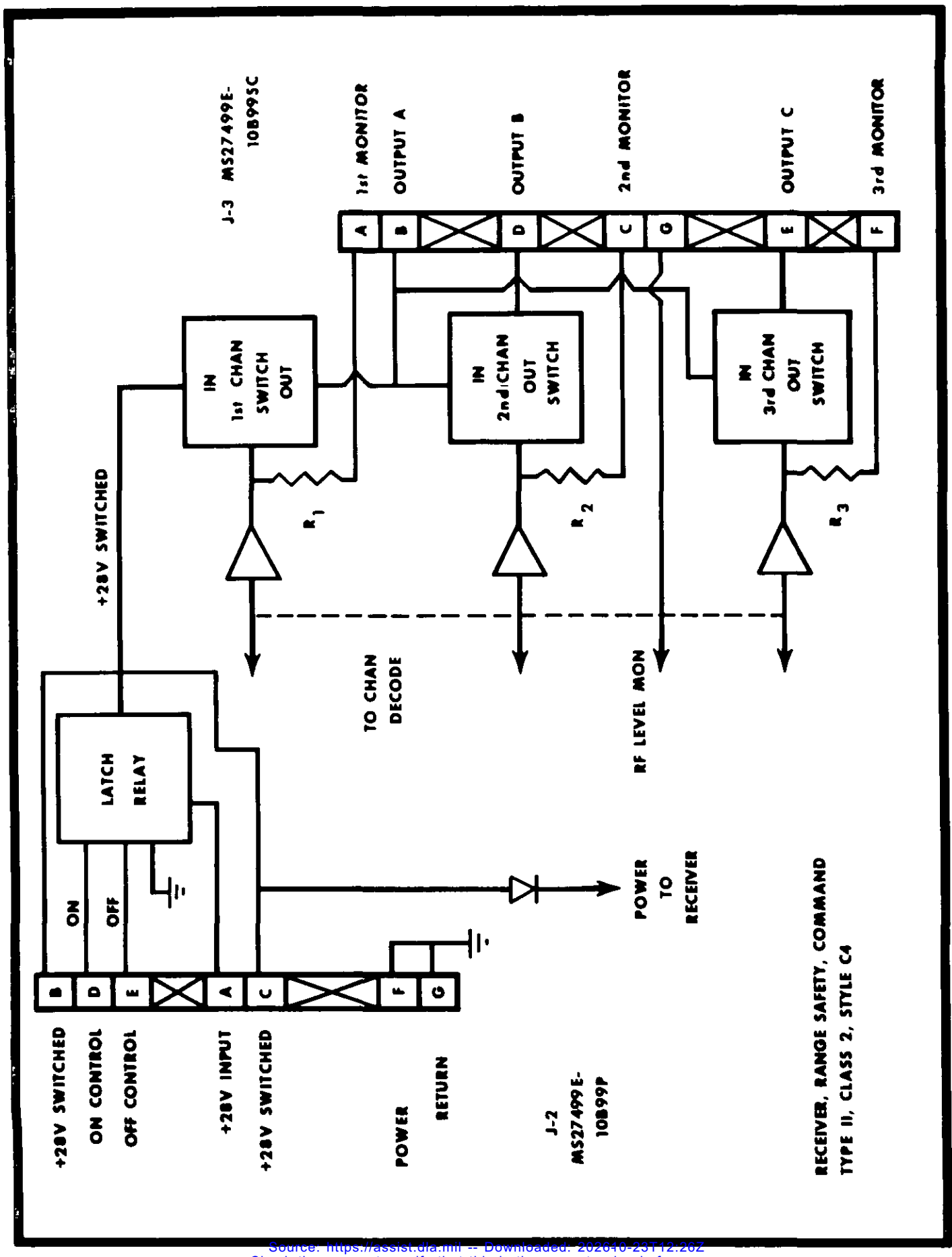


FIGURE 2D OUTPUT WIRING

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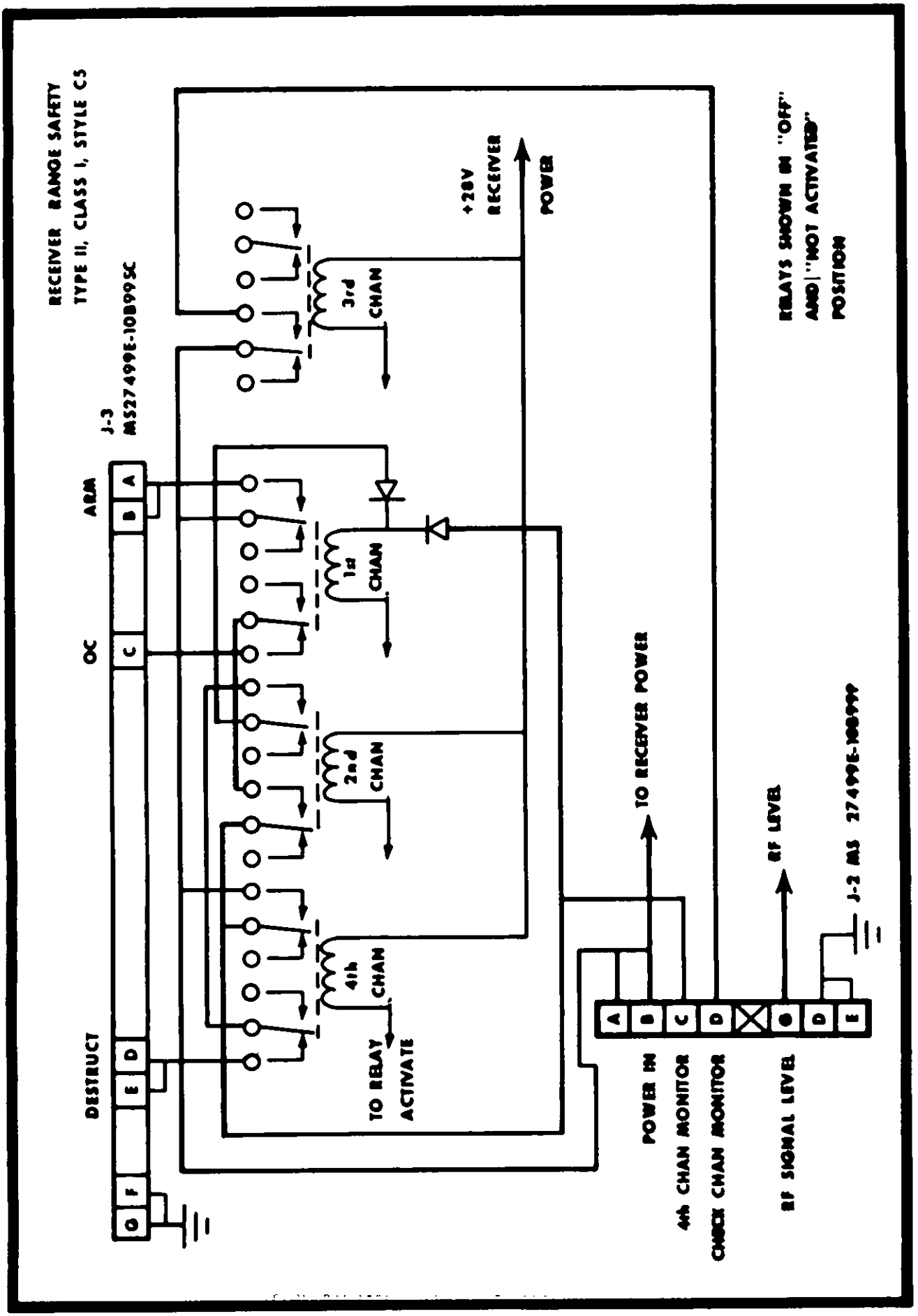


FIGURE 2. C5 CONNECTOR WIRING

FSC 1430

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